

SILTHAUL · DESIGN DECISIONS REGISTER

SiltHaul design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SLH-DEC-001	Version	0.2
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Amish's requirement decisions 2A and 7A recorded (SLH-DDR-003); boot reading for R9 opened for Amish
0.1	2026-10-03	Amish Chadha	Register opened; every decision made under Amish's pre-approval of 2026-10-03

SiltHaul design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SiltHaul's safety case (overload limit, holding, anchors, dumping). Each took the conservative option; the evidence that would relax it is in SLH-DDR-001, Table 1.

■ Open decisions

#	To decide	Options	Recommendation	What it affects in the build	Source
1	What "fits a small car boot" in R9 means. State: with the flat-pack frame the kit packs into about 454 L, 77 % of a small hatchback's load space with the rear seats folded (1,250 x 950 x 500 mm); with the seats up the 890 x 814 mm side frames do not lie flat in a typical 950 x 650 mm boot floor.	A: accept the rear seats folded as the reference and record R9 as met on paper. B: split each side frame into bolted tubes (rails, posts, braces) so everything fits with the seats up, at about forty bolts and well over 20 min of setup (R8).	A: the seats-folded space is what a volunteer's small car offers in practice, and B breaks R8.	B would replace the side frame making sketch and add about 28 bolts and their holes	SLH-DDR-003; SLH-CAL-001, L3b and L3c

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Release load of the shear pins from the bar actually bought	The 4 mm pin is sized on 240 MPa shear; a stronger bar raises the overload limit above 3,000 N	SLH-CAL-001, F; R11
2	Recommended loads of the M12 wedge anchor bought, in the concrete grade expected	The tail block check uses 4 kN tension and 6 kN shear	SLH-CAL-001, I; R7
3	Rope breaking strength and splice efficiency of the rope bought	Factor 5.4 at the limit assumes 18 kN and 90 % at the splice	SLH-CAL-001, G; R6
4	Sheave bore and rating	Pins are 25 mm; the sheaves must be rated at least 1,000 kg	BOM line 14
5	Bearing centre heights and bolt pitches	The pads and top plates are drilled to UCP206 and UCP205 catalogue sizes	SLH-DWG-101
6	Sprocket bores and the 12-tooth sprocket's fit on the 36 mm hub	Sets the hub and the line-up with the 48-tooth wheel	SLH-DWG-104

■ Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,055.60 (USD 944.40 under the target). Main cost drivers and savings worth trying:

- The two boxes (USD 160) and the winding drum (USD 98) are the largest made parts; the rope (USD 80.60), the ramp (USD 73) and the frame (USD 72) follow.
- Bought lifting gear (sling, shackles, two bridle sets, USD 164) is kept at rated, traceable grades because it carries the safety case.
- Savings worth trying: a drum from a scrap gas cylinder or pipe offcut of the same diameter; a timber ramp with a steel wear strip; buying rope with factory eye splices in bulk.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Split winding drum in place of a friction capstan; 4:1 chain drive with two cranks	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 1, 2
2026-10-03	Shear pin limits rope tension to about 2.5 kN; all parts sized on 3,000 N; requirement R11 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 3
2026-10-03	Two opposite ratchet wheels and pawls on the drum shaft; requirement R12 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 4
2026-10-03	Tail block on four drilled floor anchors only, never walls or door frames	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 5
2026-10-03	Capstan anchored by a round sling to a tree or vehicle, level or rising 10 degrees at most; stakes only against skating	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 6
2026-10-03	10 mm polyester double braid, 18 kN	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 7
2026-10-03	Box open toward the door, sloped back, forward tipping; two people on the tipping bar	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 8, 9
2026-10-03	Capstan at least 6 m beyond the door	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 10
2026-10-03	First co-design candidate to approach: Kerala State Disaster Management Authority (not agreed)	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 11
2026-10-03	SiltHaul capstan recorded as the candidate common block for SaltDrag; CalRig as the first candidate proof-load rig	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 12
2026-10-03	budget_usd kept at 2,000 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	SLH-DDR-001, item 13

Date	Decision	Decided by	Record
2026-10-03	R1 kept as the trial target and reported not met on paper (superseded by 2A below)	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, Consequences
2026-10-03	Design for construction: the thirteen changes of SLH-DDR-002	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-002
2026-10-03	Appearance model additions for renders: wound rope on the drum, stakes above ground only, context wall, tree and mannequin	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	docs/ REVIEW.md, TRL 3
2026-10-03	2A: second box on the return leg so the rope loop carries mud both ways; R1 restated as "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets" (now 0.57, met on paper)	Amish: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	SLH-DDR-003, items 1 to 5
2026-10-03	7A: bolted flat-pack frame (two welded side frames, four bolted cross members) that fits a small car boot, keeping the 25 kg heaviest-part limit (heaviest lift 23.1 kg)	Amish: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	SLH-DDR-003, items 6 to 8